

Assignment-2

DIGITAL SYSTEMS LAB

VERILOG (COMBINATIONAL LOGIC)

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QUESTION 1: 16-BIT VECTOR REVERSAL

Design Code

```
SIMULATION - Behavioral Simulation - Functional - sim_1 - testb_rev

2_Q1.v x 2_Q1.v (2) x Untitled 3 x
C:/Users/Navya/2_Q1/2_Q1.srcs/sources_1/new/2_Q1.v

1 timescale 1ns / 1ps
2
3 module rev(
4     input [15:0] in,
5     output [15:0] out
6 );
7     assign out = {in[0], in[1], in[2], in[3], in[4], in[5], in[6], in[7],
8                 in[8], in[9], in[10], in[11], in[12], in[13], in[14], in[15]};
9 endmodule
10
```

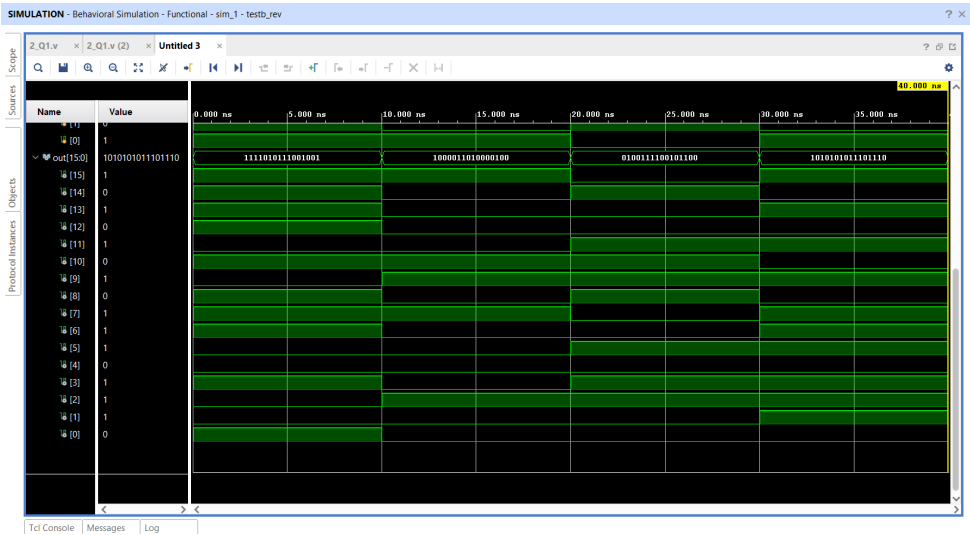
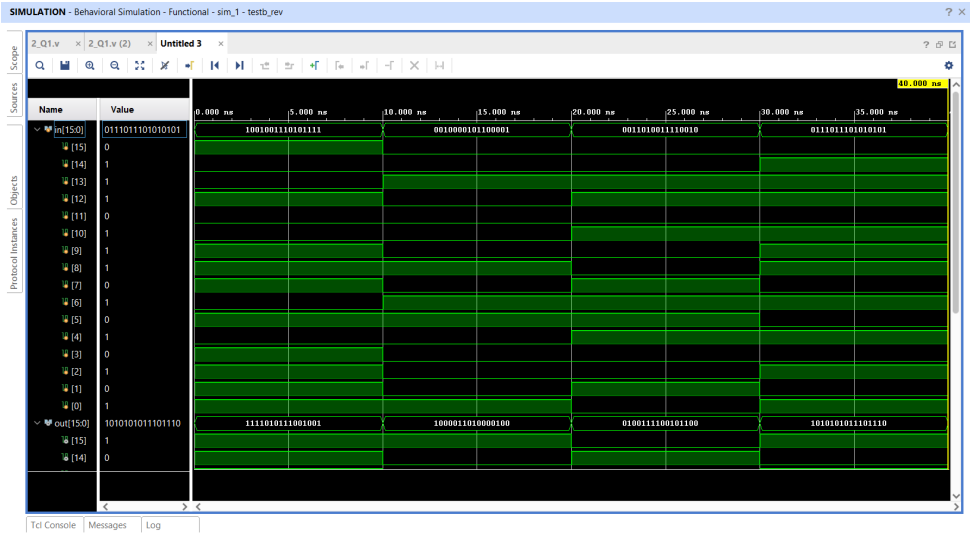
Testbench Code

```
SIMULATION - Behavioral Simulation - Functional - sim_1 - testb_rev

2_Q1.v x 2_Q1.v (2) x Untitled 3 x
C:/Users/Navya/2_Q1/2_Q1.srcs/sim_1/new/2_Q1.v

1 timescale 1ns / 1ps
2 module testb_rev;
3     reg [15:0] in;
4     wire [15:0] out;
5
6     rev uut (
7         .in(in),
8         .out(out)
9     );
10    initial begin
11        $monitor("Time=%0t | in=%b | out=%b", $time, in, out);
12        in = 16'b1001001110101111; #10;
13        in = 16'b0010000101100001; #10;
14        in = 16'b001101001110010; #10;
15        in = 16'b011011101010101; #10;
16        $finish;
17    end
18 endmodule
19
```

Simulation Waveform



Output

Scope

Sources

Tcl Console

Messages

Log

Time resolution is 1 ps

Time=0 | in=1001001110101111 | out=1111010111001001

Time=10000 | in=0010000101100001 | out=1000011010000100

Time=20000 | in=0011010011110010 | out=0100111100101100

Time=30000 | in=0111011101010101 | out=1010101011101110

\$finish called at time : 40 ns : File "C:/Users/Navya/2_Q1/2_Q1.srcs/sim_1/new/2_Q1.v" Line 16

Explanation

In this circuit, we reverse the order of bits in a 16-bit input. The first bit becomes the last and the last bit becomes the first. The simulation shows that the output is the reversed form of the input.

QUESTION 2: WIDE 2-TO-1 MULTIPLEXER

Design Code

Testbench Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - testb_mux

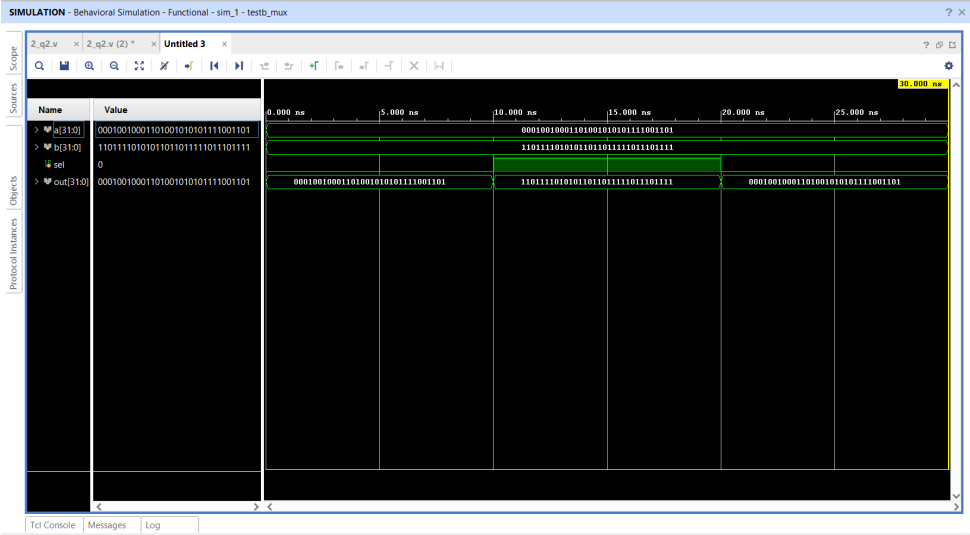
2.q2.v x 2.q2.v (2) * x Untitled 3 x

C:/Users/Navya/2_Q2/2_Q2.srscs/sim_1/new/2.q2.v

ScopeSourcesObjectsProtocol Instances

```
1 timescale 1ns / 1ps
2 module testb_mux;
3
4     reg [31:0] a, b;
5     reg sel;
6     wire [31:0] out;
7
8     mux uut (
9         .a(a),
10        .b(b),
11        .sel(sel),
12        .out(out)
13    );
14
15    initial begin
16        $monitor("Time=%0t | sel=%b | a=%d | b=%d | out=%d",
17            $time, sel, a, b, out);
18
19        a = 32'h1234_ABCD;
20        b = 32'hDEAD_BEEF;
21
22        sel = 0; #10;
23        sel = 1; #10;
24        sel = 0; #10;
25        $finish;
26    end
27 endmodule
```

Simulation Waveform



Output

SIMULATION - Behavioral Simulation - Functional - sim_1 - testb_mux

Tcl Console Messages Log x

Time resolution is 1 ps
Time=0 | sel=0 | a= 305441741 | b=3735928559 | out= 305441741
Time=10000 | sel=1 | a= 305441741 | b=3735928559 | out=3735928559
Time=20000 | sel=0 | a= 305441741 | b=3735928559 | out= 305441741
\$finish called at time : 30 ns : File "C:/Users/Navya/2_Q2/2_Q2.srcs/sim_1/new/2_q2.v" Line 26

Explanation

A 2-to-1 multiplexer selects one of two 32-bit inputs based on the select signal sel. When sel = 0, input *a* is passed to the output, and when sel = 1, input *b* is selected and so on... The waveform verifies correct switching between inputs based on sel.

QUESTION 3: MINIMUM OF FOUR NUMBERS

Design Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_min4

Scope

Sources

Objects

Protocol Instances

```
Untitled 1 x 2_33.v x 2_3.v x
C:/Users/Navya/2_Q3/2_Q3.srscs/sources_1/new/2_3.v

1  `timescale 1ns / 1ps
2  module min4(
3      input [7:0] a, b, c, d,
4      output [7:0] min
5  );
6
7      wire [7:0] m1, m2;
8
9      assign m1 = (a < b) ? a : b;
10     assign m2 = (c < d) ? c : d;
11     assign min = (m1 < m2) ? m1 : m2;
12
13 endmodule
```

Testbench Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_min4

Untitled 1 x 2_33.v x 2_3.v x

C:/Users/Navya/2_Q3/2_Q3.srscs/sim_1/new/2_33.v

Scope

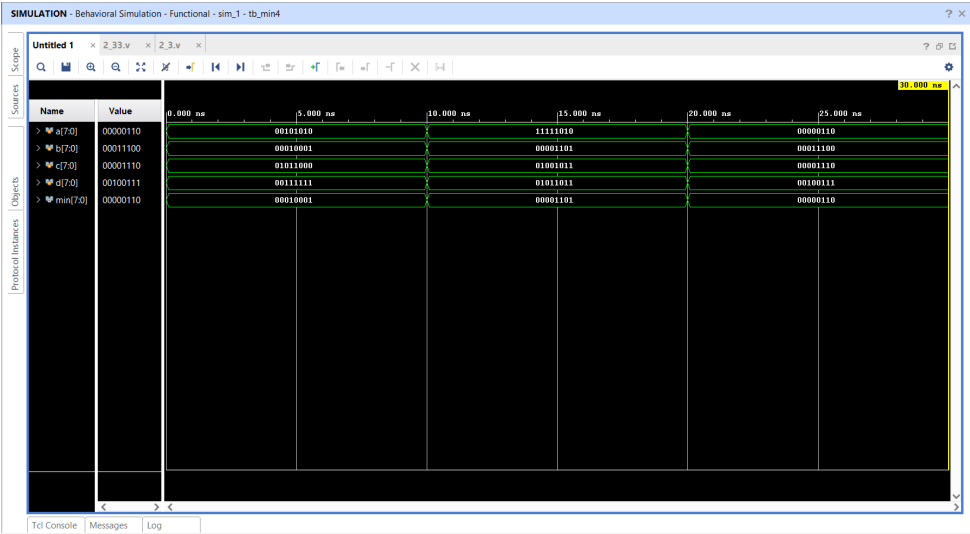
Sources

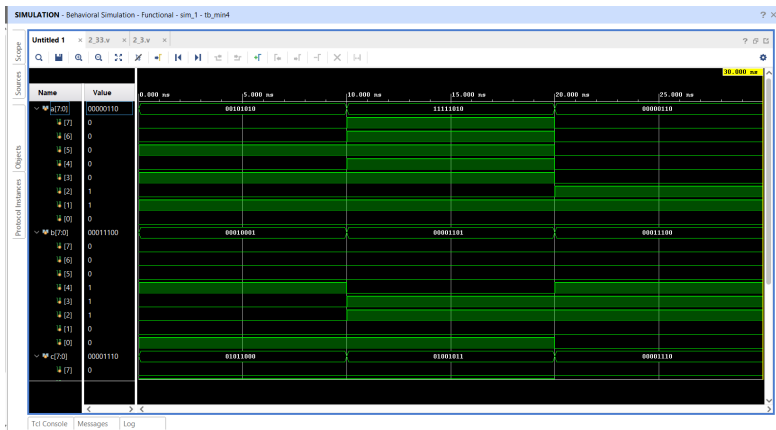
Objects

Protocol Instances

```
1  timescale 1ns / 1ps
2  module tb_min4;
3
4      reg [7:0] a, b, c, d;
5      wire [7:0] min;
6
7      min4 uut (
8          .a(a), .b(b), .c(c), .d(d),
9          .min(min)
10     );
11
12     initial begin
13         $monitor("Time=%0t | a=%d b=%d c=%d d=%d | min=%d",
14             $time, a, b, c, d, min);
15
16         a=42; b=17; c=88; d=63; #10;
17         a=250; b=13; c=75; d=91; #10;
18         a=6; b=28; c=14; d=39; #10;
19
20         $finish;
21     end
22 endmodule
```

Simulation Waveform





Output

Explanation

In this design, we find the smallest value among four 8-bit inputs by comparing them step by step. First, we compare a with b and c with d to get two smaller values. Then, we compare those two results to get the final minimum value. The simulation output shows the correct smallest value.

QUESTION 4: 3-BIT POPULATION COUNT

Design Code

Testbench Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_count

2_q4.v x 2_q44.v x Untitled 1 x

C:/Users/Navya/2_Q4/2_Q4.srcs/sim_1/new/2_q44.v

Q

1 timescale 1ns / 1ps

2 module tb_count;

3

4 reg [2:0] in;

5 wire [1:0] count;

6

7 count uut (

8 .in(in),

9 .count(count)

10);

11

12 initial begin

13 \$monitor("Time=%0t | in=%b | count=%d", \$time, in, count);

14

15 in=3'b000; #10;

16 in=3'b001; #10;

17 in=3'b010; #10;

18 in=3'b011; #10;

19 in=3'b100; #10;

20 in=3'b101; #10;

21 in=3'b110; #10;

22 in=3'b111; #10;

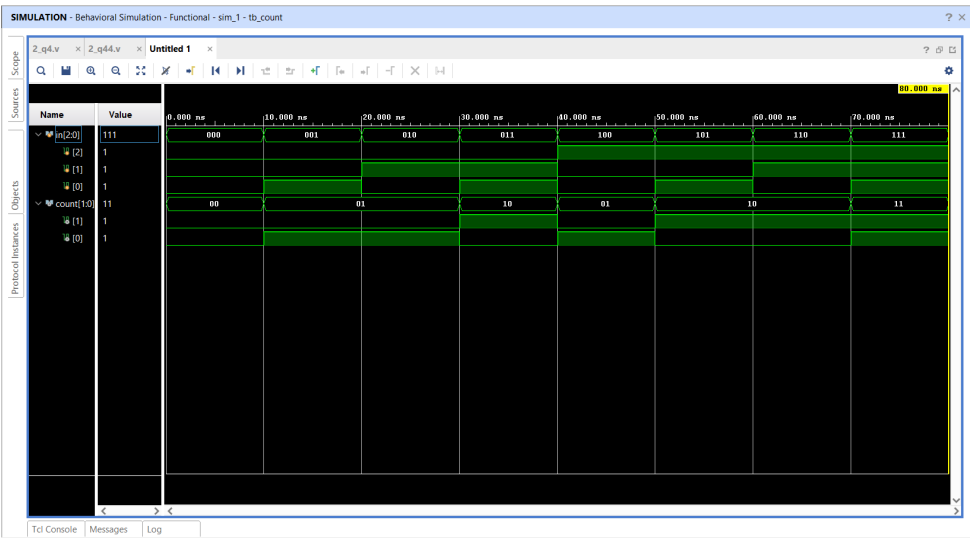
23

24 \$finish;

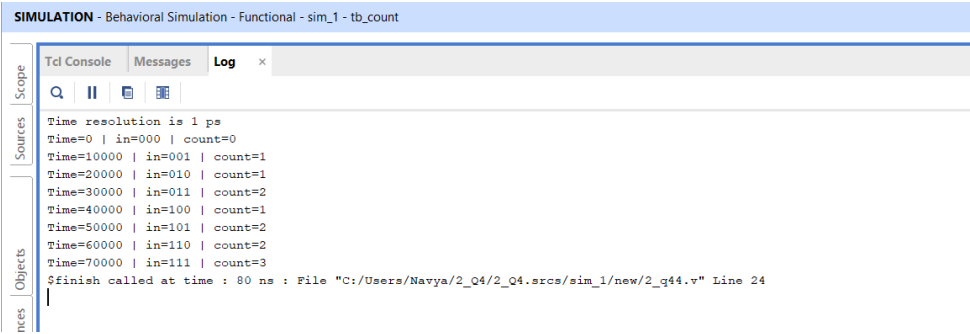
25 end

26 endmodule

Simulation Waveform



Output

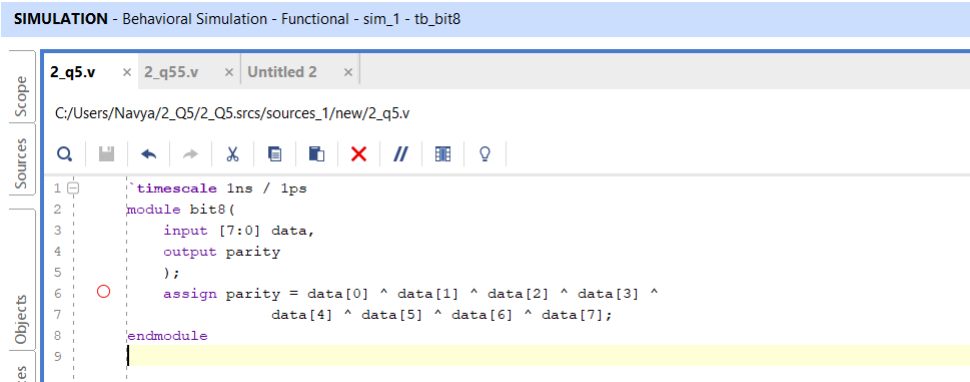


Explanation

In this circuit, we count the number of 1s present in a 3-bit input. The output shows how many bits are equal to 1. For different input combinations, the output changes from 0 to 3 accordingly. The simulation waveform confirms that the count is correct for all cases.

QUESTION 5: 8-BIT EVEN PARITY GENERATOR

Design Code



Testbench Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_bit8

2.q5.v x 2.q55.v x Untitled 2 x

C:/Users/Navya/2_Q5/2_Q5.srscs/sim_1/new/2.q55.v

ScopeSourcesObjectsProtocol Instances

1234567891011121314151617181920212223

```
1 `timescale 1ns / 1ps
2 module tb_bit8;
3
4     reg [7:0] data;
5     wire parity;
6
7     bit8 uut (
8         .data(data),
9         .parity(parity)
10    );
11
12    initial begin
13        $monitor("Time=%0t | data=%b | parity=%b", $time, data, parity);
14
15        data = 8'b10110010; #10;
16        data = 8'b00100001; #10;
17        data = 8'b10001010; #10;
18        data = 8'b11000001; #10;
19
20        $finish;
21    end
22 endmodule
23
```

Simulation Waveform

SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_bit8

2.q5.v x 2.q55.v x Untitled 2 x

ScopeSourcesObjectsProtocol Instances

0.000 ns5.000 ns10.000 ns15.000 ns20.000 ns25.000 ns30.000 ns35.000 ns40.000 ns

Name	Value
data[7:0]	111100001
[7]	1
[6]	1
[5]	1
[4]	0
[3]	0
[2]	0
[1]	0
[0]	1
parity	0

Time (ns)	data[7:0]	parity
0.000	10110010	0
10.000	00100001	0
20.000	10001010	0
30.000	11100001	0

Tcl Console | Messages | Log

11

Output

SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_bit8

Tcl ConsoleMessagesLog x

Q||

Time resolution is 1 ps
Time=0 | data=10110010 | parity=0
Time=10000 | data=00100001 | parity=0
Time=20000 | data=10001010 | parity=1
Time=30000 | data=11100001 | parity=0
\$finish called at time : 40 ns : File "C:/Users/Navya/2_Q5/2_Q5.srscs/sim_1/new/2_q55.v" Line 20

ScopeSources

its

Explanation

In this circuit, we generate an even parity bit for an 8-bit input. If the number of 1s in the input is odd, we set the parity bit to 1 to make the total count even. If the number of 1s is already even, the parity bit remains 0. The simulation waveform shows that the parity is generated correctly.

QUESTION 6: 9-TO-1 MULTIPLEXER

Design Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_mux9

2_q6.v x 2_q6.v (2) x Untitled 2 x

C:/Users/Navya/2_Q6/2_Q6.srscs/sources_1/new/2_q6.v

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ScopeSources

Objects

Protocol Instances

```
1 timescale 1ns / 1ps
2
3 module mux9(
4     input [15:0] a,b,c,d,e,f,g,h,i,
5     input [3:0] sel,
6     output [15:0] out
7 );
8     assign out = (sel==0) ? a :
9                 (sel==1) ? b :
10                (sel==2) ? c :
11                (sel==3) ? d :
12                (sel==4) ? e :
13                (sel==5) ? f :
14                (sel==6) ? g :
15                (sel==7) ? h :
16                (sel==8) ? i :
17                16'hFFFF;
18
19 endmodule
20
```

Testbench Code

SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_mux9

2.q6.v

2.q6.v (2)

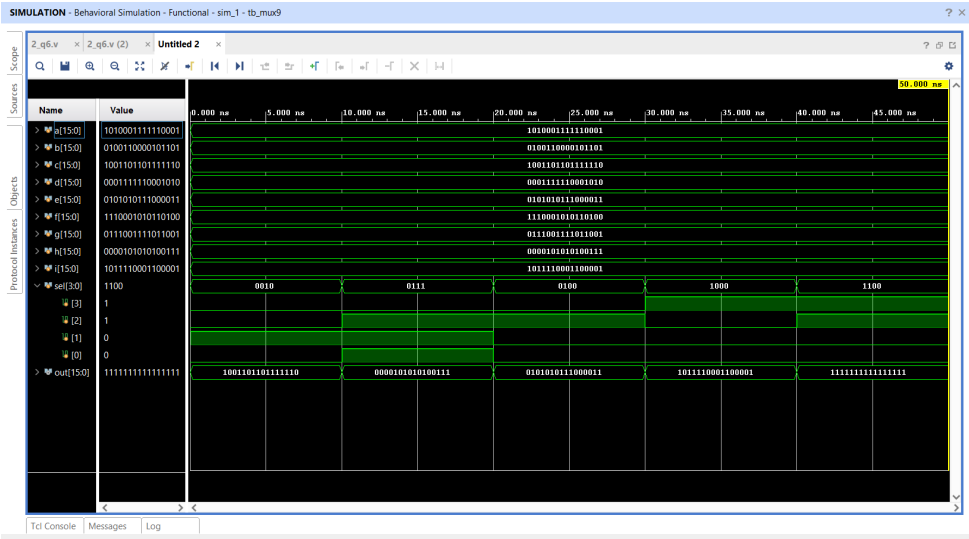
Untitled 2

C:/Users/Navya/2_Q6/2_Q6.srscs/sim_1/new/2_q6.v

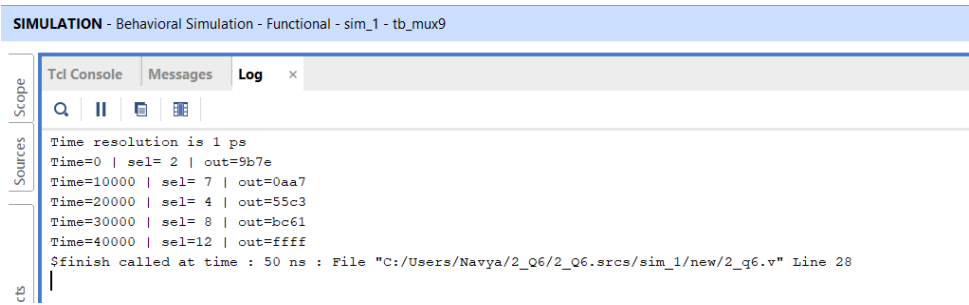
Q

```
1 timescale 1ns / 1ps
2 module tb_mux9;
3
4     reg [15:0] a,b,c,d,e,f,g,h,i;
5     reg [3:0] sel;
6     wire [15:0] out;
7
8     mux9 uut (
9         .a(a), .b(b), .c(c), .d(d),
10        .e(e), .f(f), .g(g), .h(h), .i(i),
11        .sel(sel),
12        .out(out)
13    );
14
15    initial begin
16        $monitor("Time=%0t | sel=%d | out=%h", $time, sel, out);
17
18        a=16'hA3F1; b=16'h4C2D; c=16'h9B7E;
19        d=16'h1F0A; e=16'h55C3; f=16'hE2B4;
20        g=16'h73D5; h=16'h0AA7; i=16'hBC61;
21
22        sel=2; #10;
23        sel=7; #10;
24        sel=4; #10;
25        sel=8; #10;
26        sel=12; #10;
27
28        $finish;
29    end
30 endmodule
```

Simulation Waveform



Output



Explanation

In this circuit, a 9-to-1 multiplexer is used to select one of nine 16-bit inputs based on the 4-bit select signal sel. Each value of sel from 0 to 8 selects a corresponding input. If sel is greater than 8, the output is set to all 1s. The simulation waveform shows that the correct input is selected for each value of sel.